Jacey Dai

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Education

Duke University **Aug. 2026 – May 2028**
Master in Interdisciplinary Data Science *Durham, NC*

University of Illinois Urbana-Champaign **Aug. 2022 – May 2026**
Bachelor of Science in Mathematics and Statistics, Minor in Computer Science *Champaign, IL*

- **Honors:** Dorothy E. Nowlin Scholarship Recipient, Dean's List, James Scholar

Experience

Technology Services Help Desk, UIUC **Apr. 2025 – May 2026**
Technology Support Consultant *Champaign, IL*

- Analyzed 4,000+ monthly service records using **SQL** and Excel to identify resolution bottlenecks, reducing average ticket first response time from 4 hours to 45 minutes.
- Automated weekly operational reports using **Python** and VBA, consolidating metrics across five service categories and eliminating 15 hours of manual effort weekly.
- Performed **data validation** by cross-referencing system logs with service records, identifying 500+ discrepancies and achieving 99.2% data accuracy for downstream reporting.
- Documented operational workflows with cross-functional teams, creating process guides that improved team efficiency and reduced error rates by 25%.

Illinois State Geological Survey **Nov. 2024 – Mar. 2026**
Student Assistant, Data Analytics *Champaign, IL*

- Curated **large-scale, multi-modal sensor datasets** by validating 500K+ readings using **Python (Pandas, NumPy)**, ensuring 99.5% data integrity for downstream **model training** and analysis.
- Built **automated data pipelines** using **SQL** and Python to identify, clean, and process **noisy, high-volume sensor data**, reducing processing time from 8 hours to 3.2 hours and accelerating research iteration.
- Applied **statistical anomaly-detection models (z-score and IQR-based outlier thresholding)** to flag 200+ instances of noisy or ambiguous readings, improving dataset reliability for **predictive modeling**.
- Maintained data validation protocols and documentation across multiple data sources, ensuring consistency for cross-functional research collaboration.

Illinois Risk Lab, University of Illinois **Jan. 2024 – May 2024**
Research Assistant, Data Science (Cyber Risk) *Champaign, IL*

- Built **statistical risk exposure models** using **logistic regression** and **Monte Carlo simulation** on large-scale datasets (100K+ breach records) in **Python** and SQL, achieving 87% predictive accuracy.
- Refined **model training strategies** through **hypothesis-driven benchmarking** of **logistic regression** model outputs against baseline scenarios, contributing to a framework adopted by three partner organizations.
- Conducted **P&L impact analysis** by quantifying potential loss scenarios across risk categories, informing **model refinement** and risk management strategy.
- Collaborated with faculty and industry partners to translate **statistical modeling** findings into data-driven risk-mitigation recommendations.

Power Education **May 2023 – Aug. 2023**
Data Analyst Intern *Nanjing, China*

- Analyzed 200K+ transaction records using **SQL** and **Python** to identify operational patterns, improving operational throughput by 20%.
- Automated daily reconciliation reports using **Excel VBA**, reducing manual data processing from 4 hours to 10 minutes while maintaining 99% accuracy.
- Designed an **A/B testing** framework to evaluate operational process changes, analyzing 50K+ records with 95% statistical significance to inform **experimental design** and process-optimization decisions.
- Collaborated with the Engineering team to translate analytical insights into process-automation recommendations supporting business efficiency initiatives.

Skills

- **Technical:** Python (Pandas, NumPy, SciPy, scikit-learn, statsmodels), SQL (Advanced), R, Git, Excel/VBA
- **Machine Learning & Modeling:** Logistic Regression, Regression Analysis, Monte Carlo Simulation, Statistical Anomaly Detection (z-score, IQR)
- **Statistics & Experimentation:** Hypothesis Testing, A/B Testing & Experimental Design, Risk Modeling
- **Data Engineering:** Large-Scale & Multi-Modal Data Curation, Automated Data Pipelines, Data Validation & Reconciliation, Noisy/Ambiguous Data Cleaning, Model Benchmarking
- **Communication:** Cross-Functional Collaboration, Data-Driven Storytelling, Process Documentation